



Project Architecture

Business & financial operations
Full-stack overview · Structure · Workflows

This document describes the **PathyCode Platform** — a Django + React application for digital agency operations: marketing site, client portal, quotes, PayFast payments, invoicing, project delivery, and admin finance tooling.

Executive Summary

17 Django apps	3 User tiers	JWT Authentication	PayFast Payments (ZAR)
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Technology Stack

Layer	Technologies
Backend	Django 5.2 · DRF 3.16 · SimpleJWT · django-filter · ReportLab
Frontend	React 19 · React Router 7 · Tailwind 3 · Axios · Bootstrap 5
Database	SQLite (dev) · PostgreSQL via env (production)
Payments	PayFast (redirect + ITN webhook)
Deploy	Gunicorn · CORS · python-decouple env config

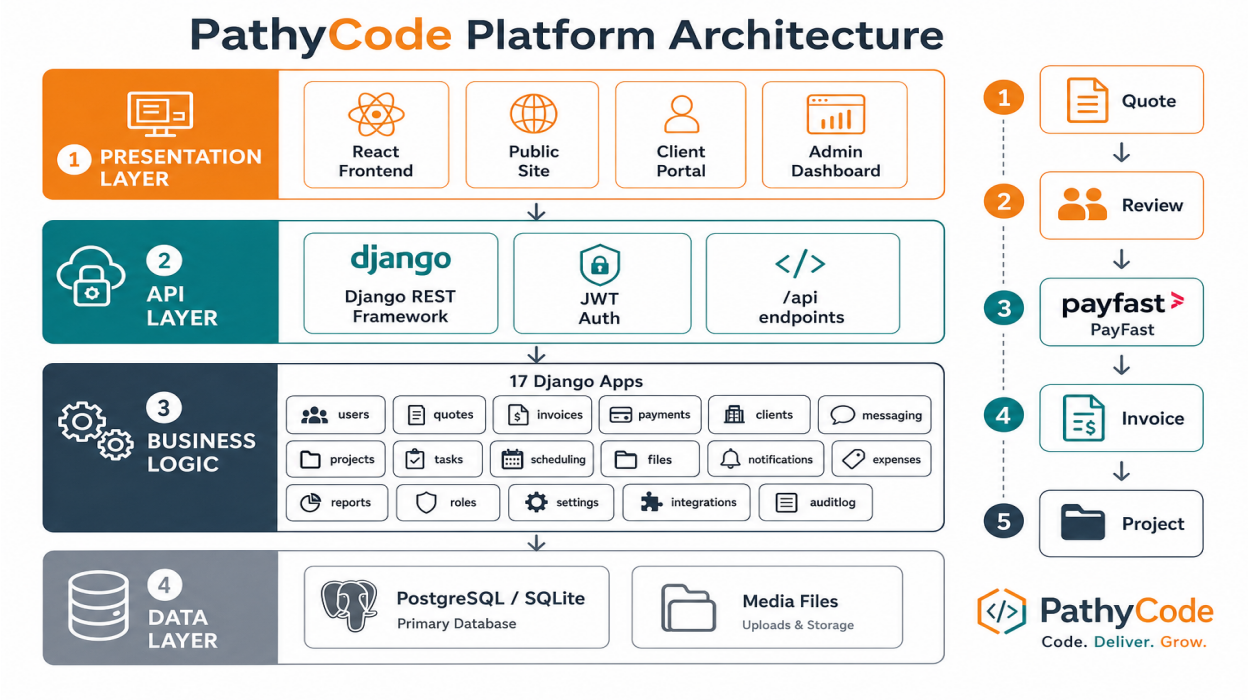
System Architecture (Layer Diagram)

Four-tier structure from browser to database:

PRESENTATION React 19 · React Router · Tailwind · Axios Public site · Client portal · Admin dashboard
API LAYER Django REST Framework · JWT auth · CORS /api/* endpoints · PayFast /payments/*
BUSINESS LOGIC 17 Django apps: quotes, invoices, clients, payments, messaging, notifications, files, tasks, calendar...
DATA SQLite (dev) / PostgreSQL (prod) · Media uploads JWT tokens · Activity & audit logs

Visual Architecture Map

High-level component and workflow structure:



Repository Structure

```
glowing-memory/  
  ■■■■ PathyCodeback/ Django project (settings, urls, csv_export)  
  ■■■■ frontend/src/  
  ■ ■■■■ pages/ Home, Profile, Admin*, Payment...  
  ■ ■■■■ components/ Navbar, SiteFooter, adminPageUi  
  ■ ■■■■ contexts/ AuthContext  
  ■ ■■■■ services/api.js Axios + JWT interceptors  
  ■■■■ users/ quotes/ invoices/ payments/ clients/ ... (17 apps)  
  ■■■■ media/ Uploaded files  
  ■■■■ docs/ Architecture docs + this PDF
```

Django Applications

App	Purpose
users	Auth, JWT, profile, activity log
clients	Client entity, paid projects, case studies, files
quotes	Quote requests & proposal workflow
invoices	Invoices, PDF export, financial dashboard
payments	PayFast gateway (redirect + ITN)
projects	Public marketing portfolio
services	Service catalog & pricing
blog	Blog posts & categories
contact	Contact form submissions
newsletter	Email subscriptions
testimonials	Client testimonials (approval)
about	About page content (singleton)
messaging	Client–admin project threads
notifications	In-app notification events
files	Shared file uploads
tasks	Work tasks on client projects
calendar_events	Calendar & reminders

Primary API Route Groups

Prefix	Description
/api/users/	Register, login, JWT, profile, password reset
/api/quotes/	Quote CRUD & workflow actions

Prefix	Description
/api/invoices/	Invoices, dashboard, PDF, mark paid
/api/payment/quote/<id>/	Payment page & PayFast start
/api/clients/	Clients, projects, tasks, case studies
/api/projects/	Portfolio showcase
/api/messaging/	Threads & messages
/api/notifications/	In-app notifications
/payments/	PayFast notify, success, cancel

Frontend Page Structure

Tier	Pages / routes
Public	Home, About, Services, Pricing, Contact. Projects, Public portfolio, Request quote. Login, Register, T
Client (auth)	Profile (hub), Client portal, Payment. My projects, Files, Tasks, Calendar. Messages, Activity log, Blo
Client (auth)	Clients list, Case studies (auth required)
Admin	Dashboard, Financial dashboard, Tasks. Quotes, Invoices, Clients, Client projects
Admin	Users, Messaging, Content (blog, services, about). Testimonials, Newsletter, Contact inbox

User Roles & Access Control

Role	Access	Key capabilities
Guest	Public pages	Browse, contact, register, view pricing
Client	Profile + portal	Quotes, pay, projects, files, messages
Staff	Partial admin	Financial dashboard, tasks
Superuser	Full /admin	All CRUD, users, quotes, invoices, content

Revenue Workflow: Quote → Payment → Project

Primary monetization path. Payment via PayFast creates an invoice and auto-starts a client project.

1	Quote submitted Client requests work — status: pending
2	Admin review Scope, price, timeline — status: reviewed
3	Client approval Client approves or rejects proposal
4	PayFast payment Online payment in ZAR
5	Invoice created Auto-generated from approved quote
6	Project started Client project created — delivery begins

Note: Two Project models exist — *clients.Project* (paid client work) and *projects.Project* (marketing portfolio). The payment workflow creates *clients.Project*.

Monetization Model

Stream	How it works	Status
Project fees	Quotes with admin-set price → PayFast payment	Live
Tiered pricing	Starter R8k+ · Professional R25k+ · Enterprise custom	Marketing
Invoicing	PDF invoices, VAT tracking, financial dashboard	Live
Maintenance	Post-launch support (manual / future recurring)	Partial
SaaS platform	Not multi-tenant — single business use	N/A

Key Integrations

Integration	Purpose
PayFast	Card/EFT payments, ITN callbacks, sandbox + production
Email (SMTP)	Password reset, notifications (console in dev)
Media storage	Local MEDIA_ROOT — logos, project images, attachments
JWT	60-min access token, 7-day refresh, auto-refresh in Axios